

BTEC Level 3 Applied Science • Unit 1 Biology • Answer sheet

Cell Structure — Answers

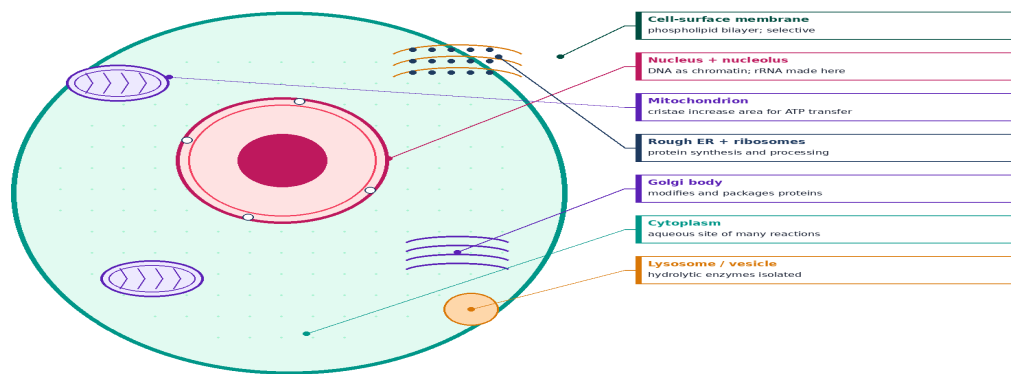
Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

Numbering matches the student worksheet exactly. Award equivalent scientifically correct wording. Show units on every calculated quantity.

A–E

1. Nucleus: DNA / control of protein synthesis. Mitochondrion: aerobic respiration / ATP. Ribosome: protein synthesis.
2. Cell wall, chloroplasts, permanent vacuole.
3. Allows some substances to cross and restricts others.
4. Animal-cell labels: cell-surface membrane, cytoplasm, nucleus / nucleolus, mitochondrion, RER + ribosomes, lysosome / vesicle. Award 1 mark per correct structure–function pair, up to 6.

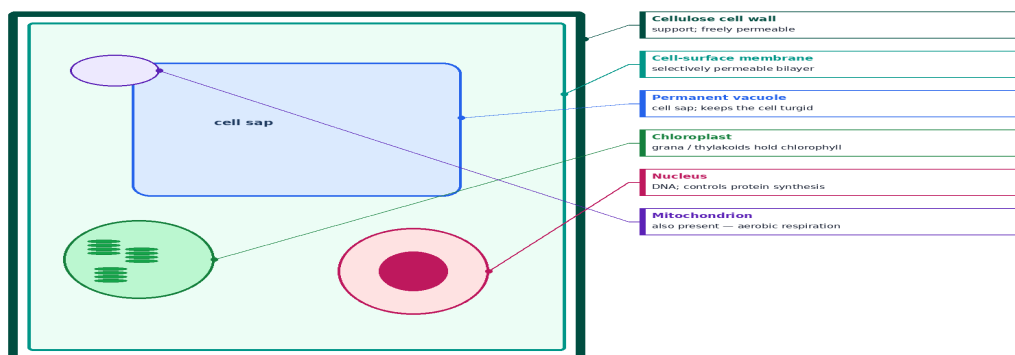
Animal cell — labelled schematic



Original schematic — organelles are not to scale. Leader lines touch the structures.

5. Plant extras: cellulose wall (freely permeable), membrane (selectively permeable), permanent vacuole, chloroplast (grana), nucleus.

Plant cell — labelled schematic



The wall is freely permeable. Exchange is controlled by the membrane inside it.

6. Phospholipid bilayer with proteins (and cholesterol / glycoproteins in animal cells); fluid mosaic.
7. Folds increase surface area for respiratory enzymes / electron-transport proteins.
8. High ATP demand for contraction.

9. Photosynthesis occurs in green shoots and leaves; roots absorb water and minerals.
10. Nucleus contains DNA and controls protein synthesis. Mitochondria carry out aerobic respiration / produce ATP. [4]
11. Many chloroplasts; position near the upper leaf; vacuole maintains turgidity; elongated shape. Any two explained. [4]
12. Wall: cellulose, freely permeable, support. Membrane: bilayer, selectively permeable. [3]
13. RER synthesises and processes proteins. Golgi modifies and packages them into vesicles for secretion.